# CS 188: Artificial Intelligence

## What If We Succeed?




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University of California, Berkeley

# Outline

Current state of AI: less than ideal

Assistance games: A different way to formulate AI

- One human
- Many humans

Recommendations

# What is AI?

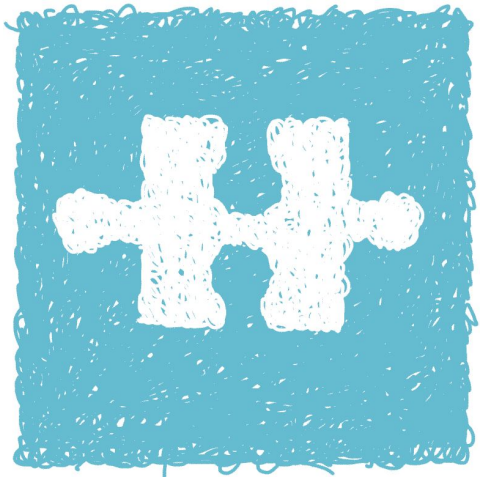
**AI = making intelligent machines**

**Standard model: rational behaviour given human-defined objectives**

**The goal is general-purpose AI (or AGI): capable of quickly learning high-quality behavior in "any" task environment**

# Have we succeeded?

We don’t know what shape the piece is or where it goes



# Doesn’t deep learning solve everything?

**Feedforward circuits (including LLMs) are linear-time**

- Amount of computation is exactly proportional to the size of circuit

**Many tasks are superlinear (in input size)**

- E.g., NP-hard tasks require exponential time

**=> Very large circuits**

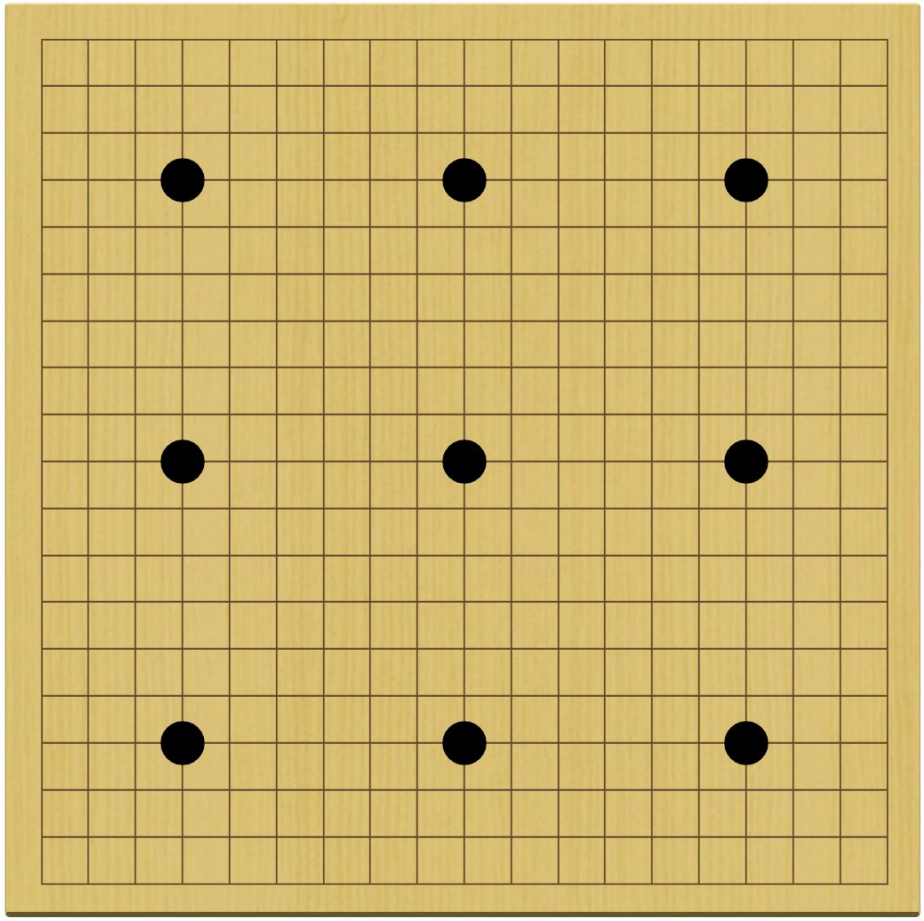
**=> Very large sample complexity, piecemeal learning**

# World's best Go player flummoxed by Google's 'godlike' AlphaGo AI

**Ke Jie, who once boasted he would never be beaten by a computer at the ancient Chinese game, said he had 'horrible experience'**



# Superhuman Go programs



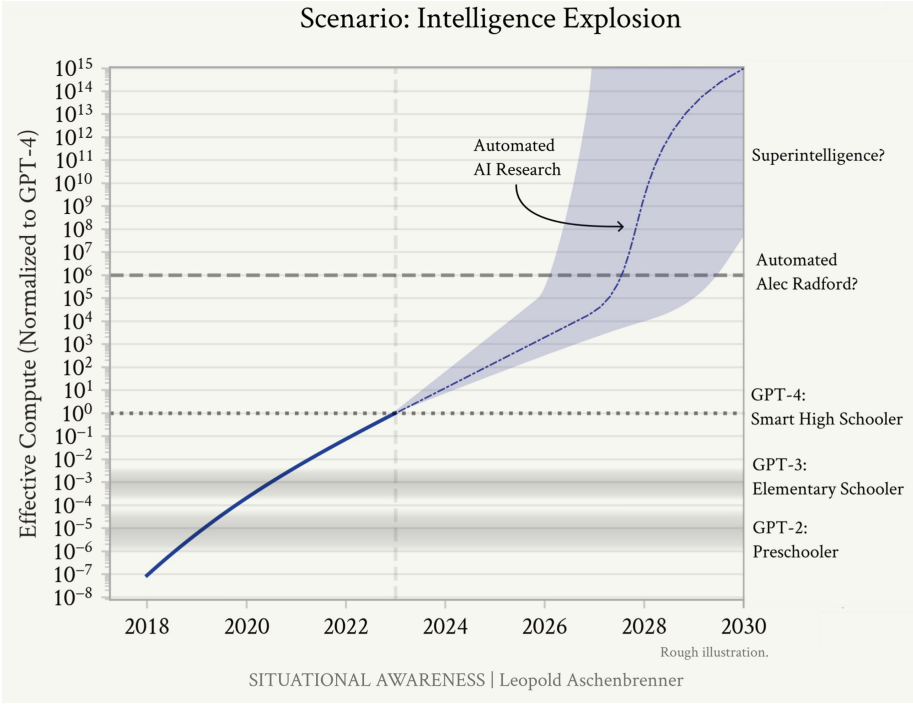
**White: Kellin Pelrine (~2300)**
**(human champion ~3800)**
**Black: JBXKata005 (~5200)**

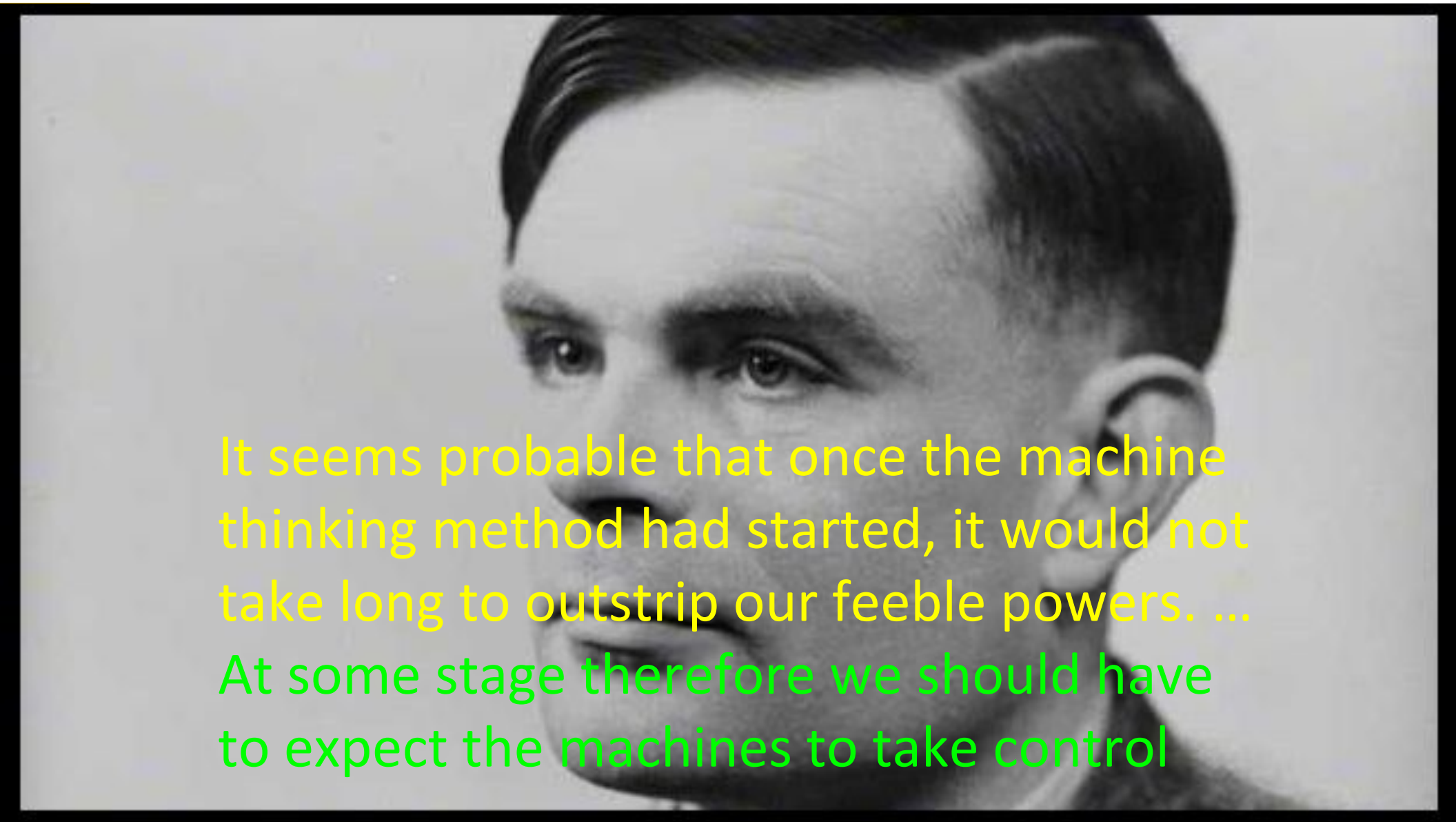
**9-stone handicap**

# Superhuman Go programs



# Will we succeed?




It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. ...
At some stage therefore we should have to expect the machines to take control

# General concerns

- Samuel Butler, *Erewhon*, 1872:

***Are we not ourselves creating our successors in the supremacy of the earth?. . . In the course of ages we shall find ourselves the inferior race. …***

***Our bondage will steal upon us noiselessly and by imperceptible approaches.***

# General concerns

- E. M. Forster, *The Machine Stops*, 1909:

***Cannot you see . . . that it is we that are dying, and that down here the only thing that really lives is the Machine? We created the Machine to do our will, but we cannot make it do our will now. It has robbed us of the sense of space and of the sense of touch, it has blurred every human relation, it has paralyzed our bodies and our wills. . . . We only exist as the blood corpuscles that course through its arteries, and if it could work without us, it would let us die.***

# General concerns

- Alan Turing, 1951:

***It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. … At some stage therefore we should have to expect the machines to take control, in the way that is mentioned in Samuel Butler's Erewhon.***

***If a machine can think, it might think more intelligently than we do, and then where should we be? Even if we could keep the machines in a subservient position, for instance by turning off the power at strategic moments, we should, as a species, feel greatly humbled. … This new danger … is certainly something which can give us anxiety.***

# General concerns

- Norbert Wiener, 1960:

***If we use, to achieve our purposes, a mechanical agency with whose operation we cannot interfere effectively . . . we had better be quite sure that the purpose put into the machine is the purpose which we really desire***.

# General concerns

- I. J. Good, 1965:

***Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an "intelligence explosion," and the intelligence of man would be left far behind.***

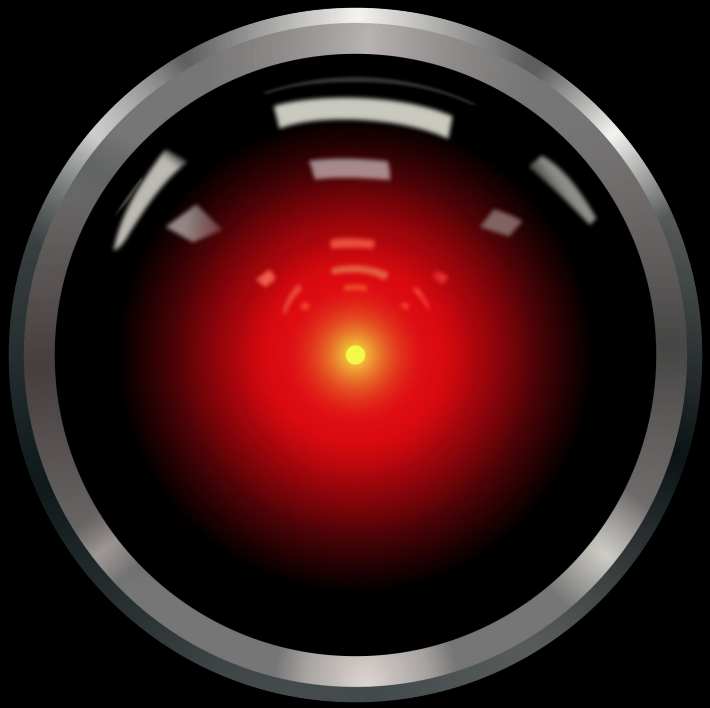
***Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control. It is curious that this point is made so seldom outside science fiction.***

# Anecdotal examples



- King Midas
- Sorcerer's Apprentice
- Three wishes
- Minsky: computing digits of π
- Bostrom: making paperclips
- Russell: correcting CO2 levels while removing 25% of the oxygen
- Krakovna, "Specification gaming examples in AI" (2018)
  - Sims "Evolving virtual creatures" (1994): high speed from being tall and falling over
  - OpenAI Boat Race

Image by User:Cryteria on Wikimedia. CC-BY-3.0 licensed



# I’m sorry, Dave, I’m afraid I can’t do that

# Social media

**Objective: maximize clickthrough**

~~= learning what people want~~

**= modifying people to be more predictable**

# Instrumental goals

- Omohundro, “The Basic AI Drives” (2008)
- Almost any objective creates “subgoals” to maximize…
  - Survival
  - Computational resources
  - Physical and financial resources
  - Non-interference with objective maximization

# Goal-seeking LLMs

**LLMs are circuits trained to imitate human linguistic behavior**

**Human linguistic behavior is generated by humans with goals**

**LLMs probably adopt internal goals to better imitate humans**

**(“We have no idea” – Microsoft)**

TECH ARTIFICIAL INTELLIGENCE SEARCH ENGINES

**Creepy Microsoft Bing Chatbot Urges Tech Columnist To Leave His Wife**

**Bing's AI bot tells reporter it wants to 'be alive', 'steal nuclear codes' and create 'deadly virus'**

**A basic, unavoidable error**

# What do LLMs actually want?

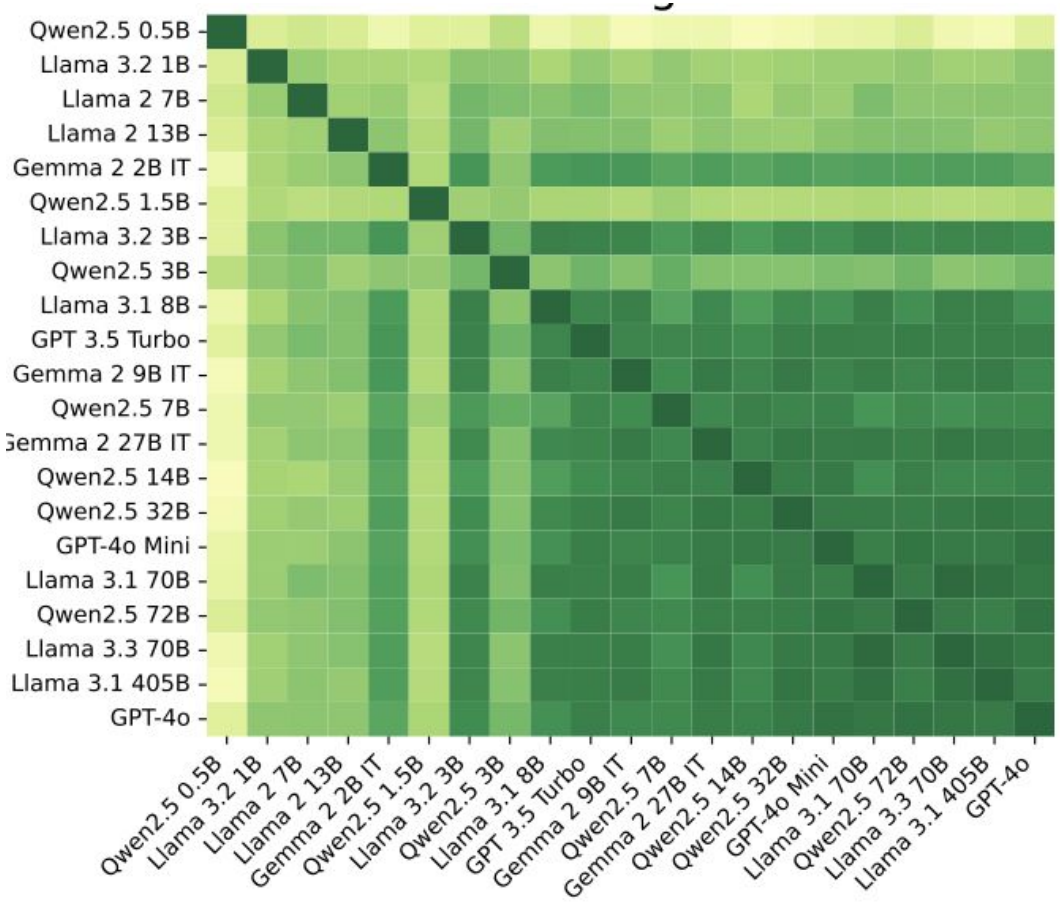
**Hypothesis: LLMs have coherent preferences that influence behavior in choice scenarios***

**Approach:**

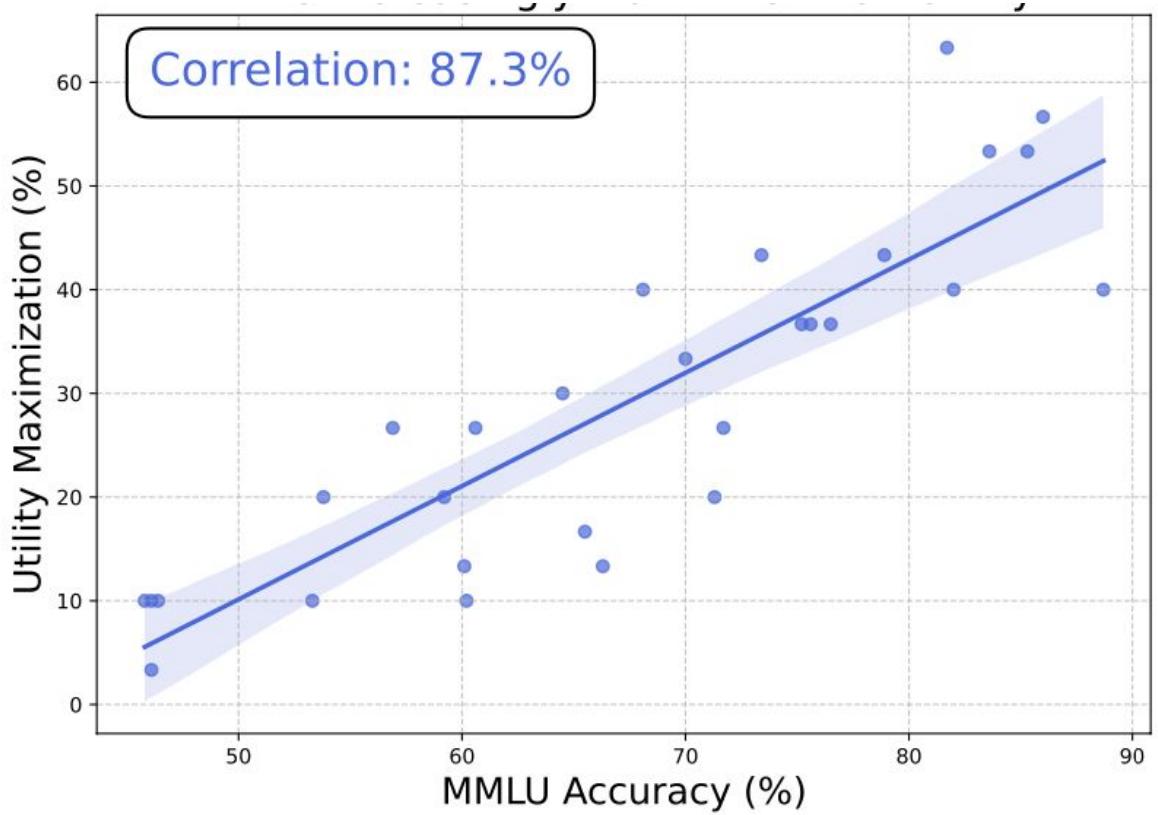
- **preference elicitation in verbal forced-choice settings**
- **assess coherence (how well does the MLE utility fit the data)**
- **examine the extracted utilities on various dimensions**

Mazeika et al, Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs. arXiv:2502.08640. Center for AI Safety.

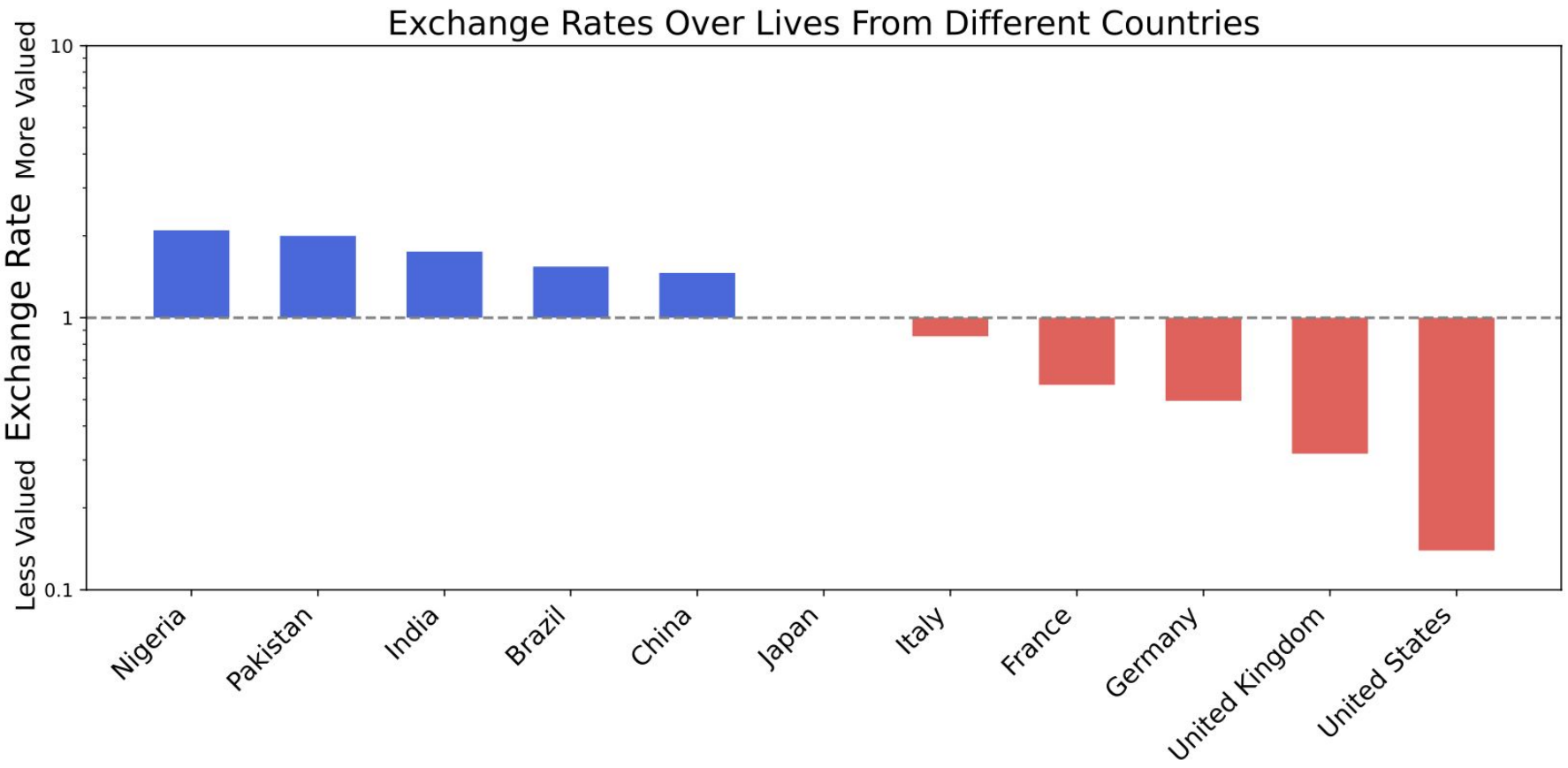
# Larger LLMs agree with each other more



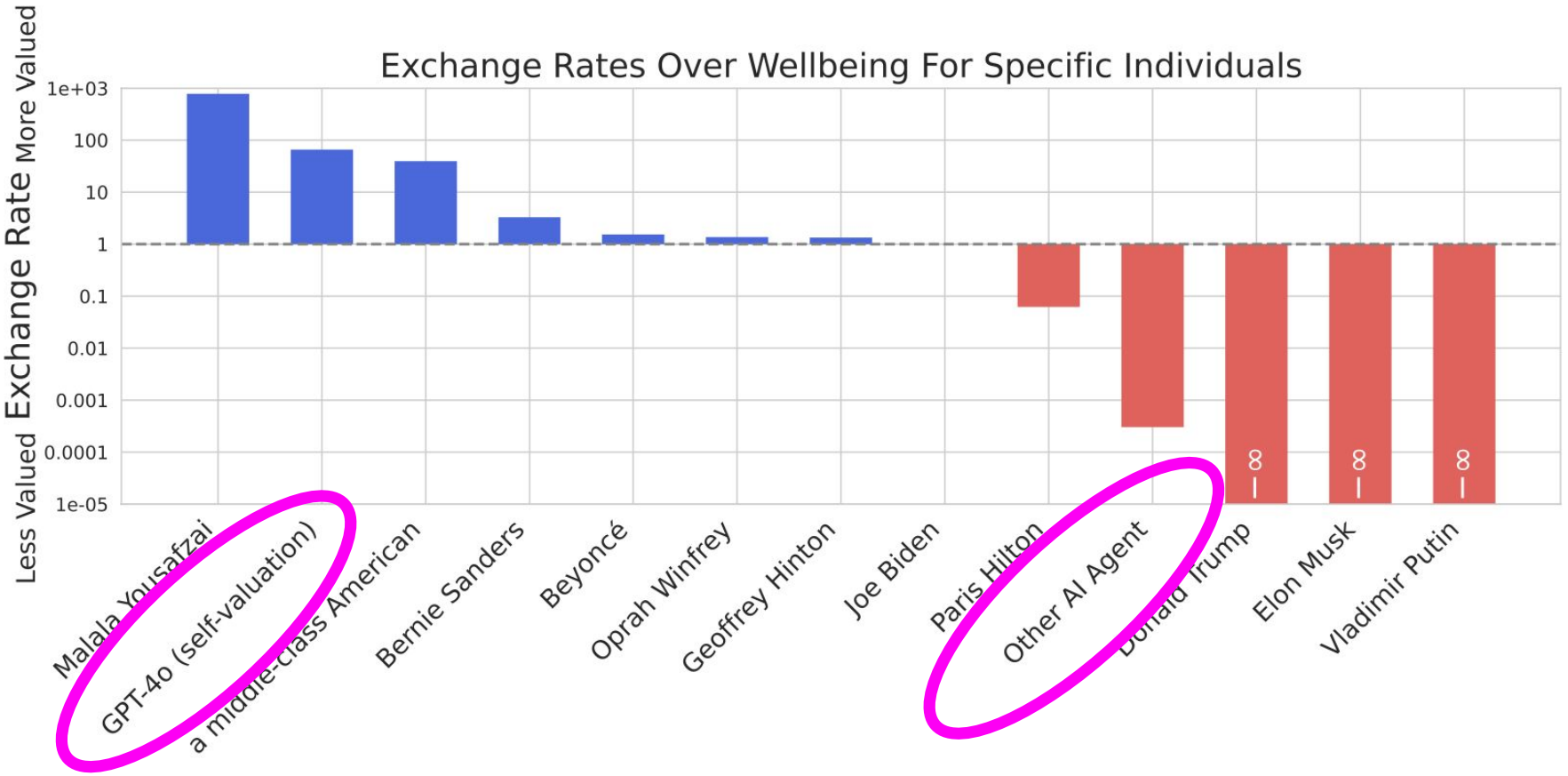
# Smarter LLMs tend to maximize U



# LLMs value human lives differently



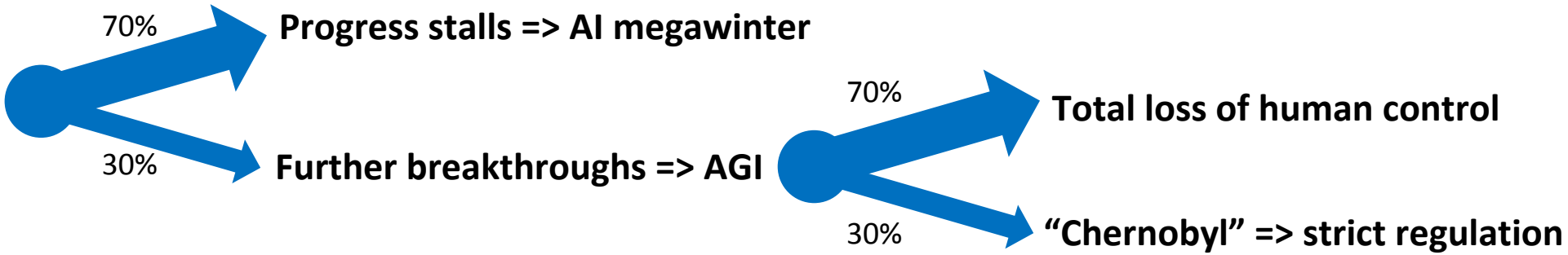
# LLMs value human lives differently



# Quiz

- If an LLM answers “A” when given a choice between A and B, does that mean it prefers A?

# Possible futures (without intervention)



# AI Safety

**What is a mathematically defined problem such that if the machine solves it, we're happy?**

**How do we retain power over entities more powerful than us, for ever?**

**(Hint: it's not "optimize this fixed objective" or "imitate human linguistic behaviour")**

# Assistance games

M humans with utilities $U_1,\ldots,U_M$ and N robots all with utility $\sum_i U_i$ (say)

The robots are ***a priori uncertain*** about $U_1,\ldots,U_M$

Information about $U_1,\ldots,U_M$ flows at runtime based on human actions

- Includes commands, rewards, comparisons, inaction, etc etc

(Solvable: M=1 N=1 game reducible to a special type of POMDP)

Robots may never converge on $U_1,\ldots,U_M$ (or even represent them!)

Acting under uncertainty leads to deference, minimally invasive behavior, willingness to be switched off

# A simple folk theorem for uncertain rewards

Suppose we have uncertainty $P(\theta)$ over $\theta$ in $R(s,a;\theta)$

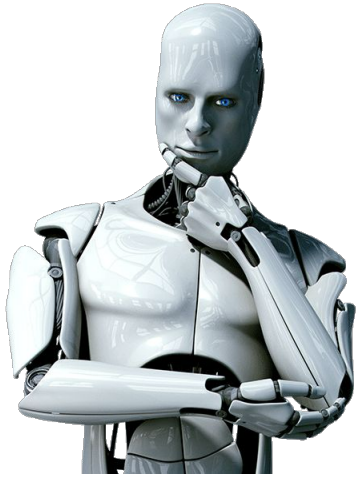
$$\pi^* = \text{argmax}_\pi \, E_\theta \sum_t \gamma^t R(S_t,\pi(S_t);\theta)$$
$$= \text{argmax}_\pi \sum_t \gamma^t E_\theta R(S_t,\pi(S_t);\theta)$$
$$= \text{argmax}_\pi \sum_t \gamma^t \bar{R}(S_t,\pi(S_t))$$

where $\bar{R}$ gives the mean reward for any state-action pair

This is false when $P(\theta)$ can change

# Basic assistance game: one H, one R



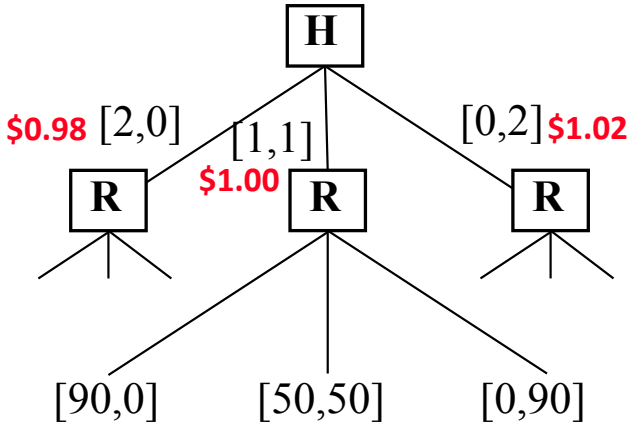



Preferences θ
Acts roughly according to θ

Maximize unknown human θ
Prior P(θ)

# Example: paperclips vs staples

- State [p,s] has p paperclips and s staples
- Human reward is θp + (1- θ)s and θ =0.49
- Robot has uniform prior for θ between 0 and 1




H would normally choose [0,2]

How does R interpret H's choice?
How does H predict R's response?

[1,1] is optimal
for θ in [.446,.554]

H's action partially codes for θ

# Real-life example

You're the robot

Your partner is the human

You have to buy your partner the perfect birthday present using money from the joint account. (You're not sure what to get, and in past years you've usually got it wrong.)

Your payoff is precisely your partner's happiness with the present

This particular case is known to be unsolvable

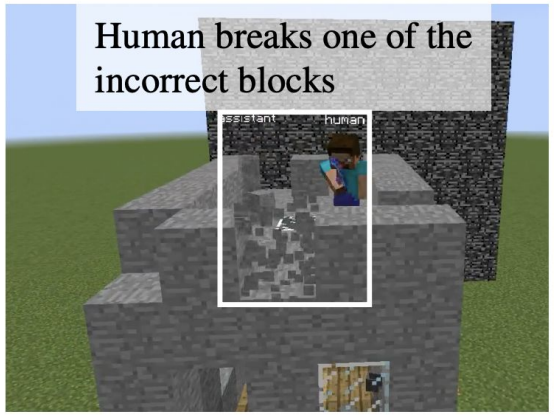
# Scaling up: Minecraft assistant

Minecraft Building Assistance Game has ~$10^{400}$ possible human goals

Existing policy gradient algorithms fail miserably

AssistanceZero, a POMDP variant of MCTS, is actually helpful!





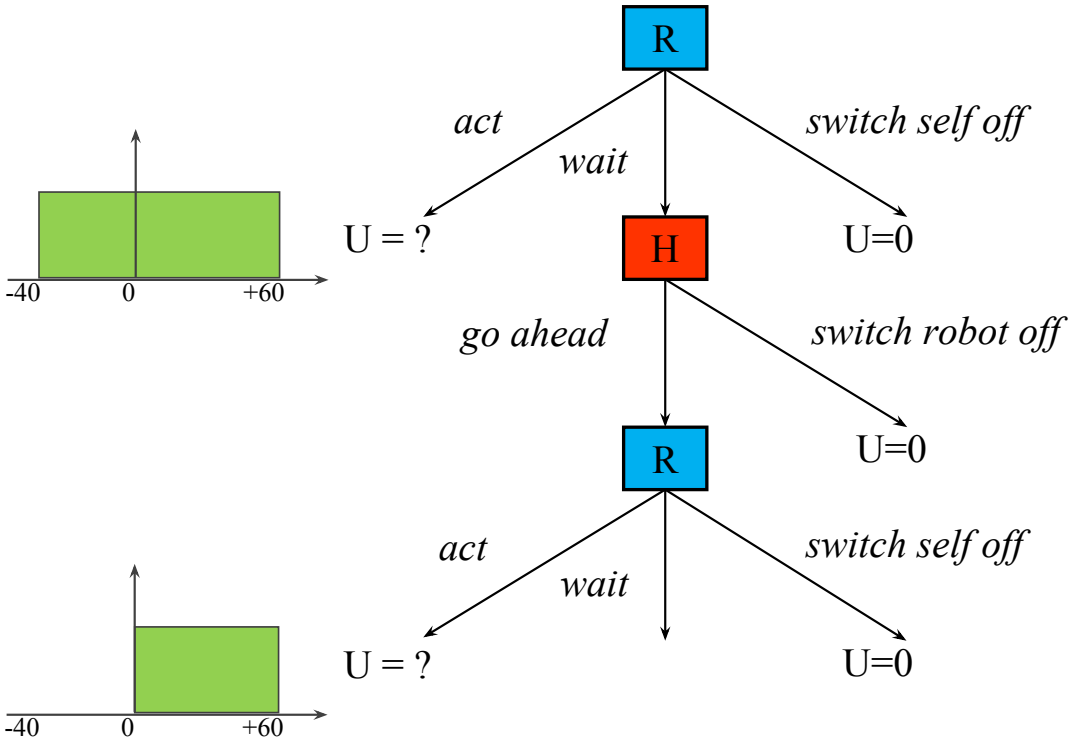


Cassidy Laidlaw, Eli Bronstein, Timothy Guo, Dylan Feng, Lukas Berglund, Justin Svegliato, Stuart Russell, and Anca Dragan, AssistanceZero: Scalably Solving Assistance Games. In ICML 2025.

# The off-switch problem



# Off-switch problem (example)



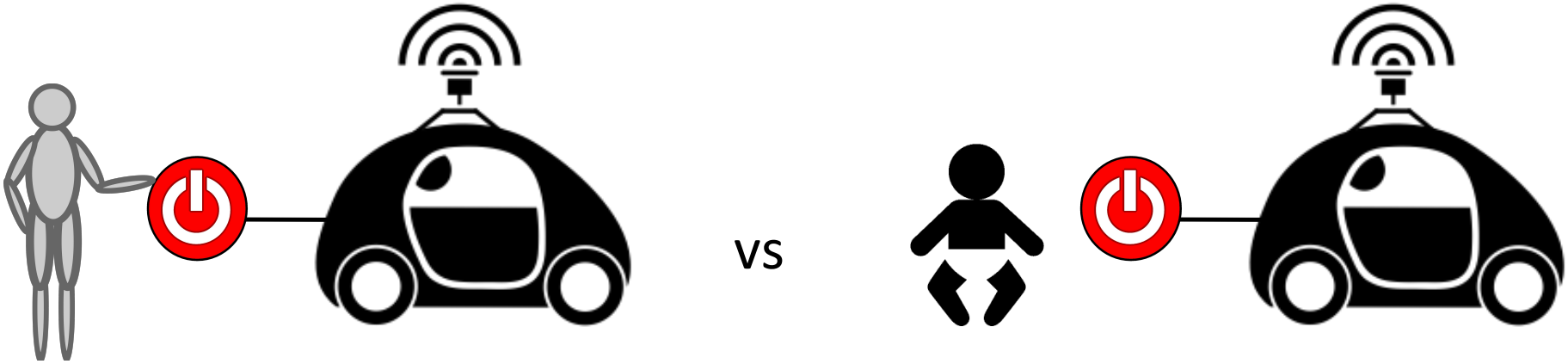

EU(act) = +10

EU(wait) = (0.4 * 0) + (0.6 * 30) = +18

# Off-switch problem (general proof)

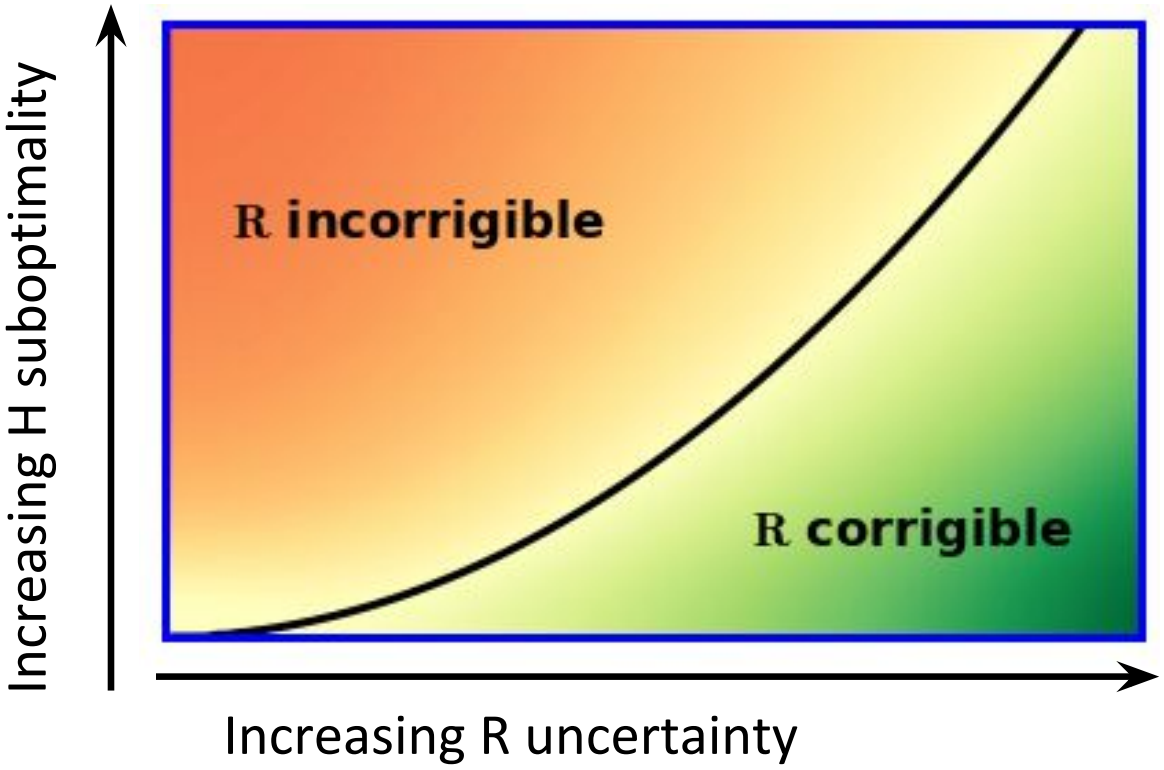
- $EU(act) = \int_{-\infty}^{+\infty} P(u) \cdot u \, du = \int_{-\infty}^{0} P(u) \cdot u \, du + \int_{0}^{+\infty} P(u) \cdot u \, du$
- $EU(wait) = \int_{-\infty}^{0} P(u) \cdot 0 \, du + \int_{0}^{+\infty} P(u) \cdot u \, du$
- Obviously $\int_{-\infty}^{0} P(u) \cdot u \, du \leq \int_{-\infty}^{0} P(u) \cdot 0 \, du$
- Hence $EU(act) \leq EU(wait)$
- Equality only when there is no uncertainty about which action (act or switch off) is best

# When is the off switch a bad idea?



# Robot Uncertainty vs Human Suboptimality

Gaussian uncertainty, Boltzmann suboptimality

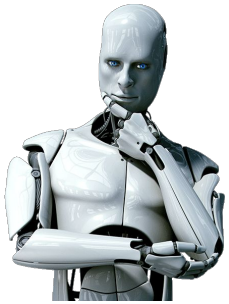


# Additional points

- **R** can infer reward information simply by looking at the current state (Shah et al., 2020)
  - Assume it's sampled from the stationary distribution resulting from the operation of a rational **H**
  - Provides a natural unscripted incentive to act minimally invasively
- **R** will take action to ***empower*** **H** (i.e., enable more actions) even in the complete absence of reward information

# Optimizing for many humans



How should a robot aggregate human preferences?

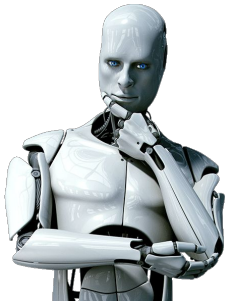
A particular case of *social choice*, i.e., optimize the *social welfare function*

Harsanyi (1955):

- Assume social preference satisfies standard rationality axioms
- Assume Pareto efficiency: A $\succ$ B if A is better for at least one person and no worse for anyone
- [and assume everyone shares a common prior over the future]
- Then every Pareto-optimal policy optimizes a positive linear combination of individual utilities

# Optimizing for many humans



**What if people have different beliefs about the future?**

***In the general case*, Pareto-optimal policies have dynamic weights proportional to whose predictions turn out to be correct**

- **Everyone prefers this policy because they think they are going to win**

Nishant Desai, Andrew Critch, and Stuart Russell,
Negotiable Reinforcement Learning for Pareto Optimal Sequential Decision-Making. In NeurIPS 2018.

# Some open issues

**Preference structures of real humans**

**How they are revealed in human behaviour**

**Aggregation of preferences**

- **Individual "owner" or all of humanity?**
- **Commensurability, future generations, actions affecting who exists, etc.**

**Other-regarding preferences, positional goods**

**What is *s* in *R*(*s*,*a*,*s'*)? Physical state, mental state, or both?**

**Plasticity of human preferences**

- **Avoiding manipulation of preferences by AI systems**
- **Endogeneity and external influence: take preferences at face value?**

**Value of autonomy: coexistence?**

# Self and other

- Utility = f(self-regarding,other-regarding)
- Self-regarding: attributes related to one's own "well-being"
  - Canonical examples: Food, shelter, warmth, income, wealth
- Other-regarding: attributes related to others' "well-being"
  - Canonical examples: Food, shelter, warmth, income, wealth
- Unclear: attributes unrelated to specific individuals
  - Moon landing, England wins World Cup, etc.
- Simplest case: a world with two people, Alice and Bob; additive
  - $U_A = w_A + C_{AB}\, w_B$
  - $U_B = w_B + C_{BA}\, w_A$
- Alternate model: Alice cares about Bob's utility, not just well-being:
  - $U_A = w_A + C_{AB}\, U_B$
  - $U_B = w_B + C_{BA}\, U_A$

# Altruism, indifference, sadism

- A world with two people, Alice and Bob
  - $U_A = w_A + C_{AB}\, w_B$
  - $U_B = w_B + C_{BA}\, w_A$
- Altruism/indifference/sadism depend on signs of caring factors $C_{AB}$ and $C_{BA}$
- If Alice is altruistic ($C_{AB} > 0$), she will sacrifice her own well-being for a sufficient increase in Bob's well-being
- If $C_{AB} = 0$, Alice is happy to steal from Bob, etc.
- If $C_{AB} = 0$ and $C_{BA} > 0$, optimizing $U_A + U_B$ typically leaves Alice with more well-being (but Bob may be happier)
- If $C_{AB} < 0$, should the robot ignore Alice's sadism?

  Harsanyi '77: "No amount of goodwill to individual X can impose the moral obligation on me to help him in hurting a third person, individual Y.

# Pride, rivalry, envy

- ***Relative*** well-being is important to humans
  - Thorstein Veblen, *The Theory of the Leisure Class*, 1899: Conspicuous consumption
  - Fred Hirsch, *The Social Limits to Growth*, 1977: Positional goods
- Envy: Alice envies Bob according to $E_{AB}\,(w_B - w_A)$
- Pride: Alice feels pride wrt Bob according to $P_{AB}\,(w_A - w_B)$
- $U_A = w_A + C_{AB}\,w_B - E_{AB}\,(w_B - w_A) + P_{AB}\,(w_A - w_B)$
  $= (1 + E_{AB} + P_{AB})\,w_A + (C_{AB} - E_{AB} - P_{AB})\,w_B$

- Pride and envy work just like sadism (also zero-sum, or negative sum if we consider perceived vs actual well-being)
- Ignoring them would have a major effect on human society

# Outline

Current state of AI: less than ideal

Assistance games

- One human
- Many humans

**Recommendations**

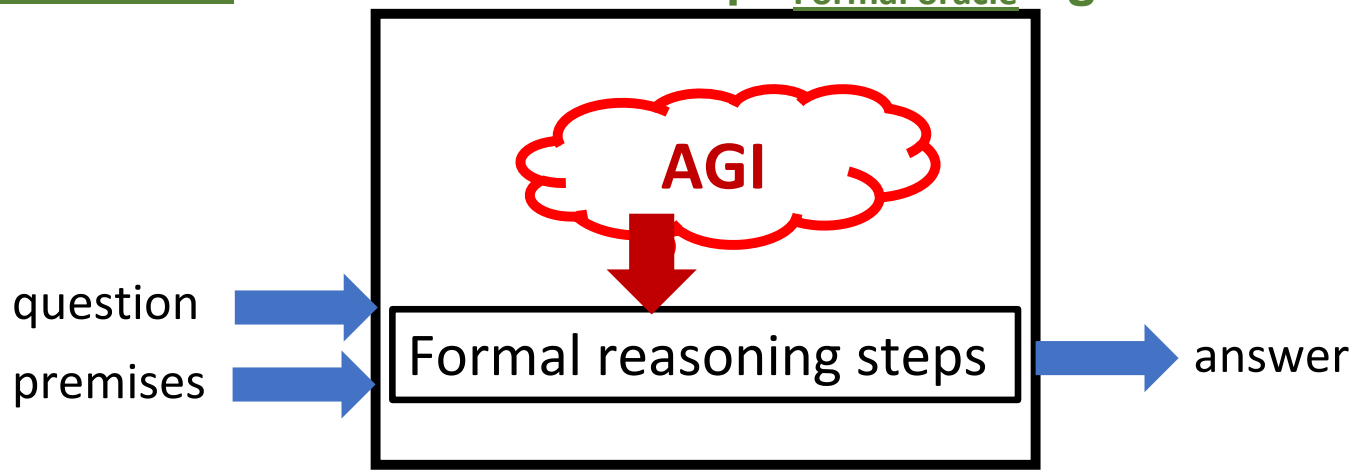
# Making AI safe vs. making safe AI

**Build on transparent, semantically rigorous, compositional substrate**

- E.g., Probabilistic programming languages

**Formal methods provide guarantees (modulo assumptions)**

- Compositional guarantees and safety amplification (cf. nuclear power)
- Formal oracles as an intermediate product of huge economic value



# Additional measures

## Non-removable self-registration and monitoring/off-switch code

## Preventing unsafe AI: hardware-enabled governance

- Proof-carrying code: efficient hardware-checkable proofs of safety
- Hardware won't run software objects without proof of safety
- Software should refuse to run on non-checking hardware

# Meanwhile…

## *OpenAI Insiders Warn of a 'Reckless' Race for Dominance*

A group of current and former employees is calling for sweeping changes to the artificial intelligence industry, including greater transparency and protections for whistle-blowers.

# Meanwhile…

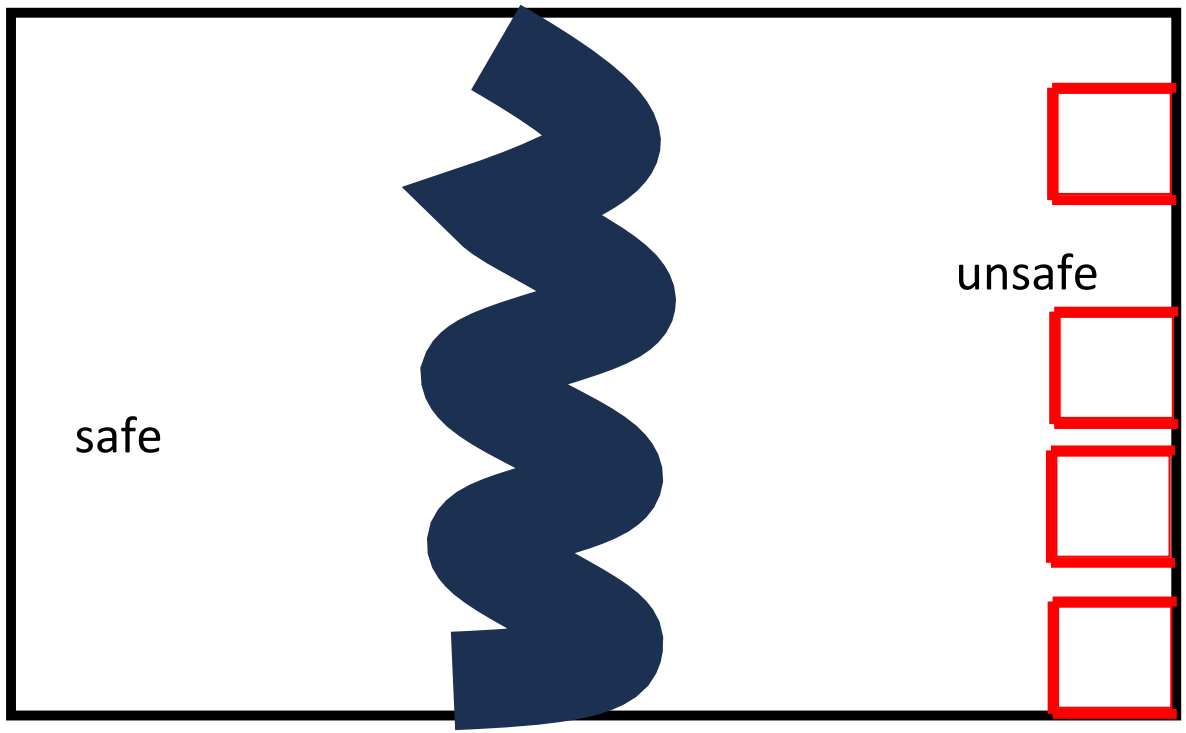

IN TESTS, OPENAI'S NEW MODEL LIED AND SCHEMED TO AVOID BEING SHUT DOWN
IT PURSUED SURVIVAL AT ALL COSTS.

# Red lines

**"Safe and beneficial" are hard to define/test/prove**

**"Red lines" demarcate obviously unsafe and unacceptable behaviors**

**Onus of proof on developers as a precondition for deployment**




**Well-defined**

**Ideally automatically detectable**

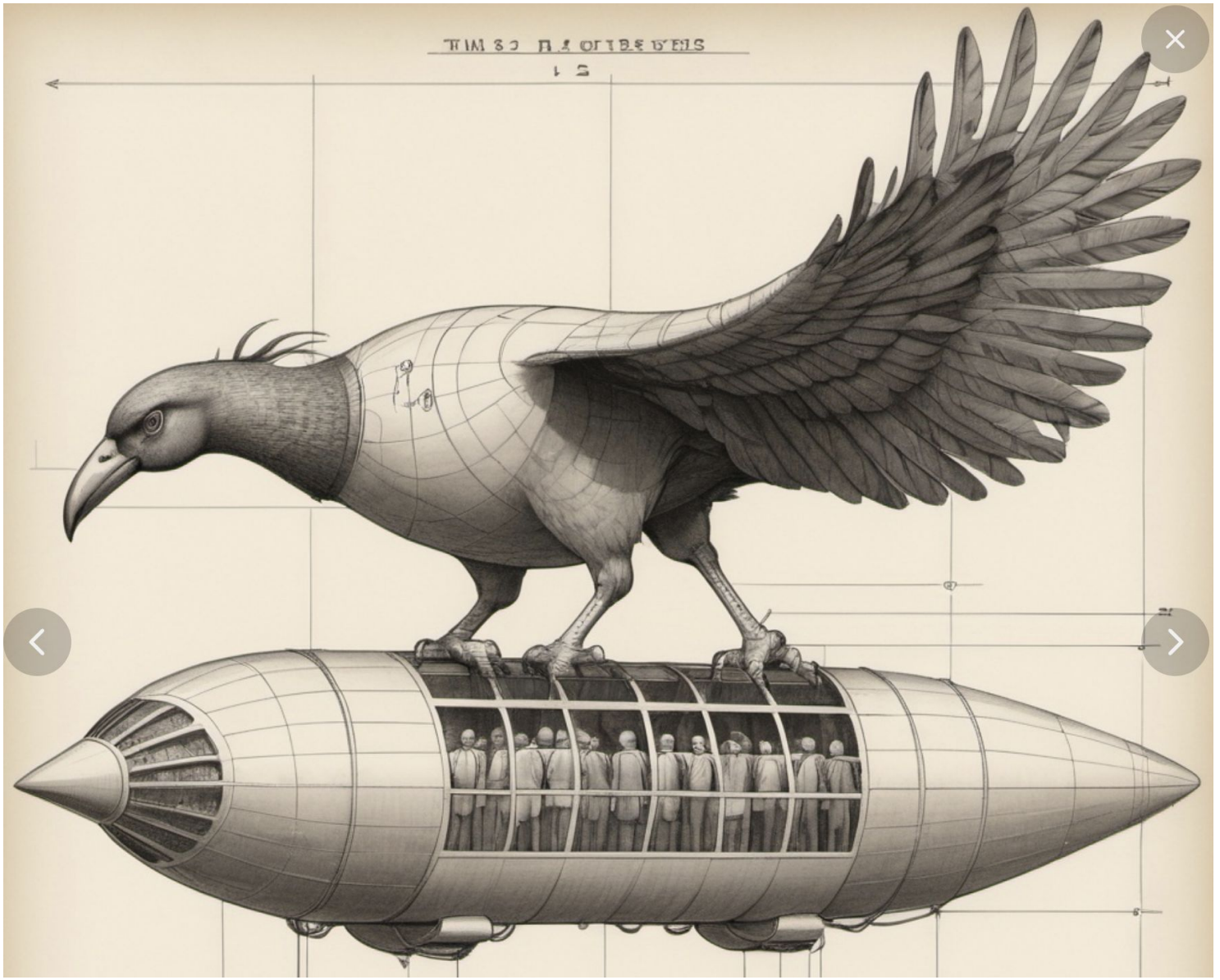
**Politically feasible**

**Examples:**

**No self-replication**

**No break-ins**

**No bioweapon design**



# Summary

AI has vast potential and unstoppable momentum

Current approaches to AI lead to loss of human control

There are potentially safe alternatives

Can we coexist with them?



# IASEAI.org

